WSU AIRPACT PROJECT UPDATE & ANNUAL REPORT

Jun Meng

with contribution from:
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and other NW-AIRQUEST members

NW-AIRQUEST Consortium Annual Meeting, July 9, 2026

Ongoing Tasks

AIRPACT5 is still running on old Aeolus HPC

Ongoing Tasks
1. Provide daily forecasts of air quality for the three-state region that can be used as guidance in daily air quality management.
2. Provide near real-time verification statistics of the current AIRPACT system accuracy by pollutant and month, as well as provide for user access to AIRPACT results and monitoring data for user-defined periods for download and analysis.
3. Track and continue to improve model performance so that model predictions provide useful guidance for air quality forecasting.
4. Continue improvements and requested additions to AQ websites.
5. Maintain the "Change Log" that is hosted at GitHub: https://github.com/wsular/airpact
6. Archive daily model output of the first 24 hours of forecast for each day and provide access to same upon requests from users. Gridded model output files should remain accessible for up to three years.
7. Modify AIRPACT as needed to respond to changes in fire emissions produced by other organizations.
8. Modeling infrastructure maintenance and updates. The updates should be optimized to maintain or improve the reliability of the system, and to enable reliable performance and, if possible, earlier delivery of the daily products.

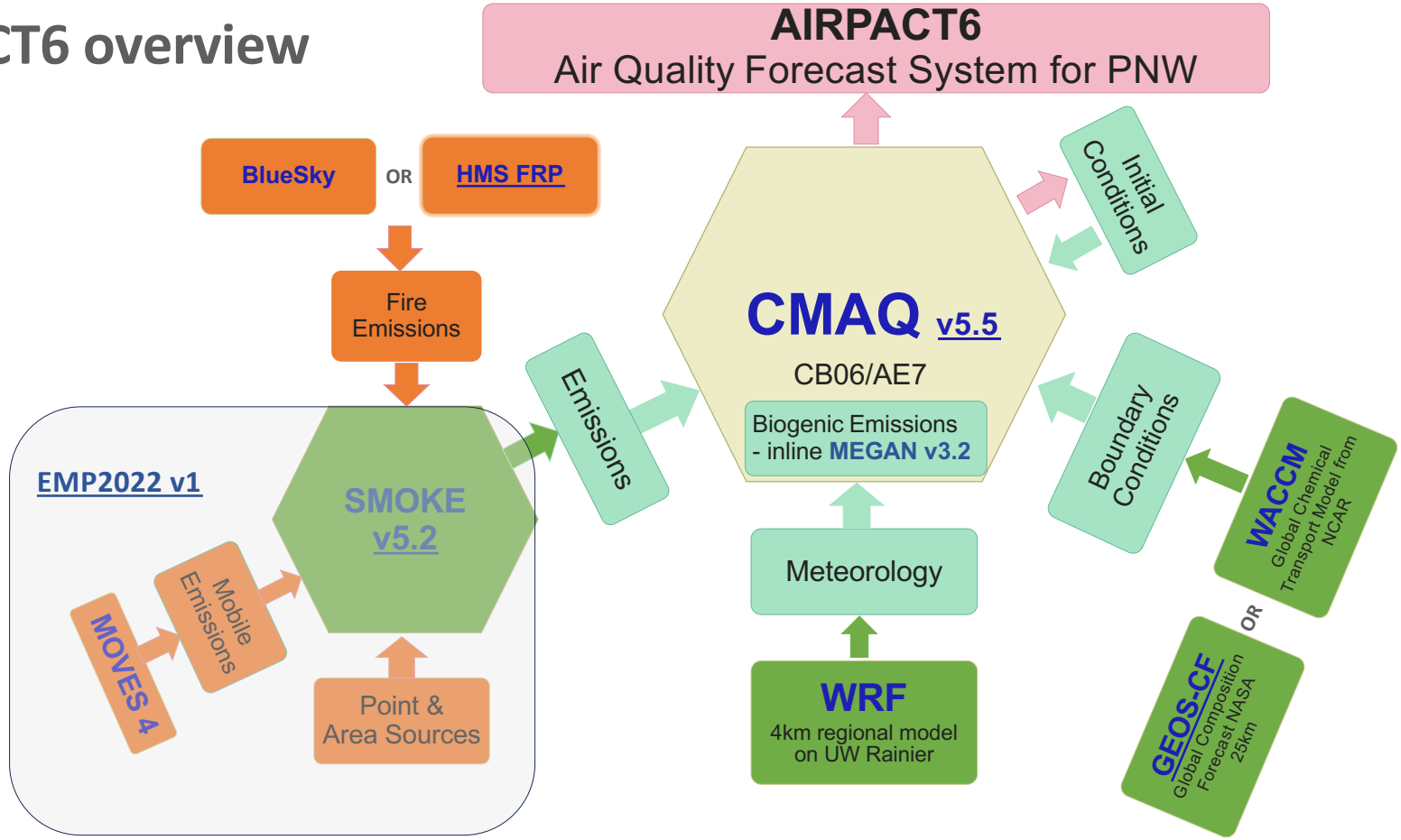
- No major issues with ongoing tasks
- AIRPACT sunset plan
- Incorporate ongoing tasks into AIRPACT6 Operation

Credits to Jahidul Islam for maintaining the AIRPACT5 forecasting system

FY26 Deliverables

1. AIRPACT6 is running forecast on WSU Kamiak HPC

AIRPACT6 overview



AIRPACT SYSTEM UPGRADE

Credits to Priom Zarrah

Upgrading to New Systems

CMAQ v5.5

Model

- Upgrade from the AIRPACT-5 CMAQ release
- CB6r5 gas-phase chemistry + AE7 aerosol module
- Updated photochemistry & deposition science

SMOKE v5.2

Emissions processing

- Runs on EPA's 2022v1 modeling platform
- Inline fire, point emissions MEGAN inline biogenic

Model Setup

Chemical Mechanism
CB6R5/AE7

Biogenic Emissions
MEGAN v3 (Inline)

Emission Inventory
EMP 2022v1 (SMOKE v5.2)

Domain
285×258 cells, 4-km, LCC

Boundary Conditions
WACCM

Meteorology
WRF - MCIP

HPC Configuration

3
Compute Nodes

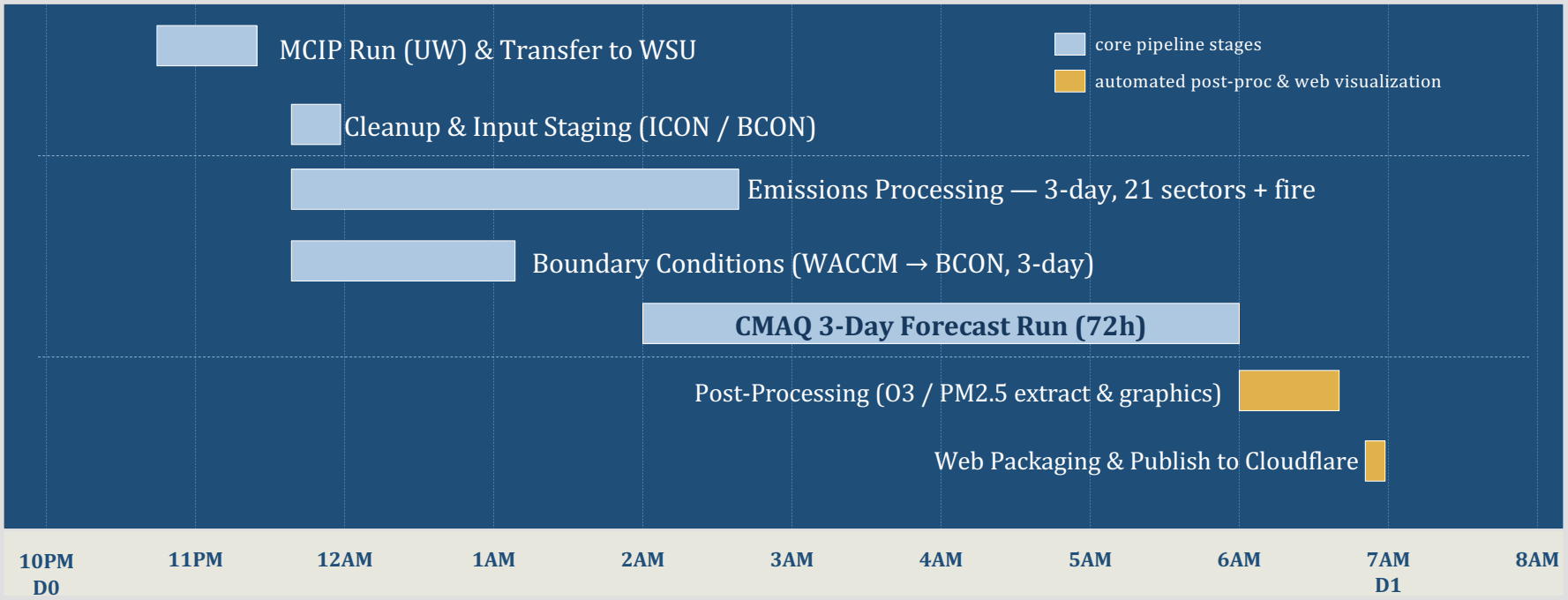
64
CPUs per Node

~2 hrs/day
Wall Time

AIRPACT-6 OPERATION

AIRPACT5 operation needs around 11 hours; while AIRPACT6 needs around 8 hours

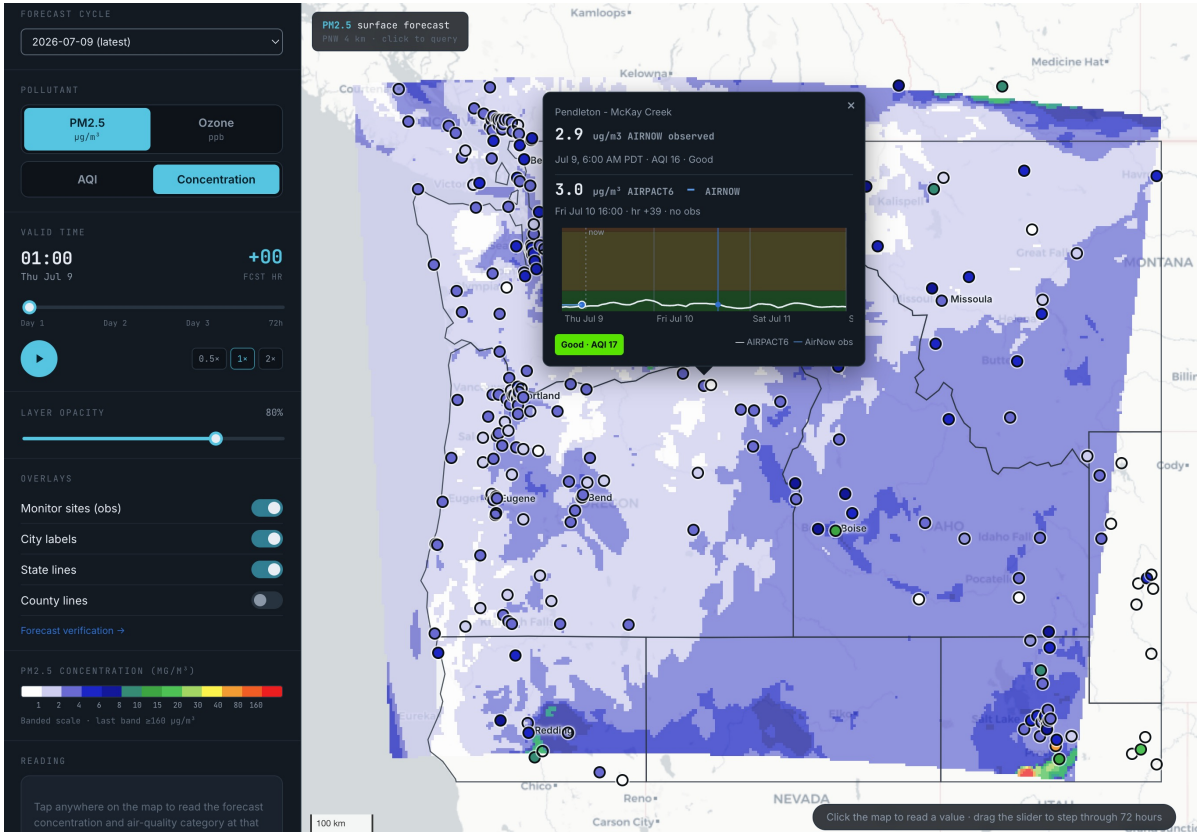
- 3-day forecast (D1–D3): single 72h CMAQ v5.5 run starting from 10:40pm nightly
- automated post-processing & web visualization (Cloudflare for now)
- Operation Schedule (nightly cycle, Pacific Time):



Operation schedule (Pacific Time). Bar lengths are current SLURM time allocations with 3 full nodes — real run-times are less.

2. AIRPACT6 visualization Prototype is now live

Interactive Map Prototype - https://nw-air-forecast.pages.dev/



[Demo]

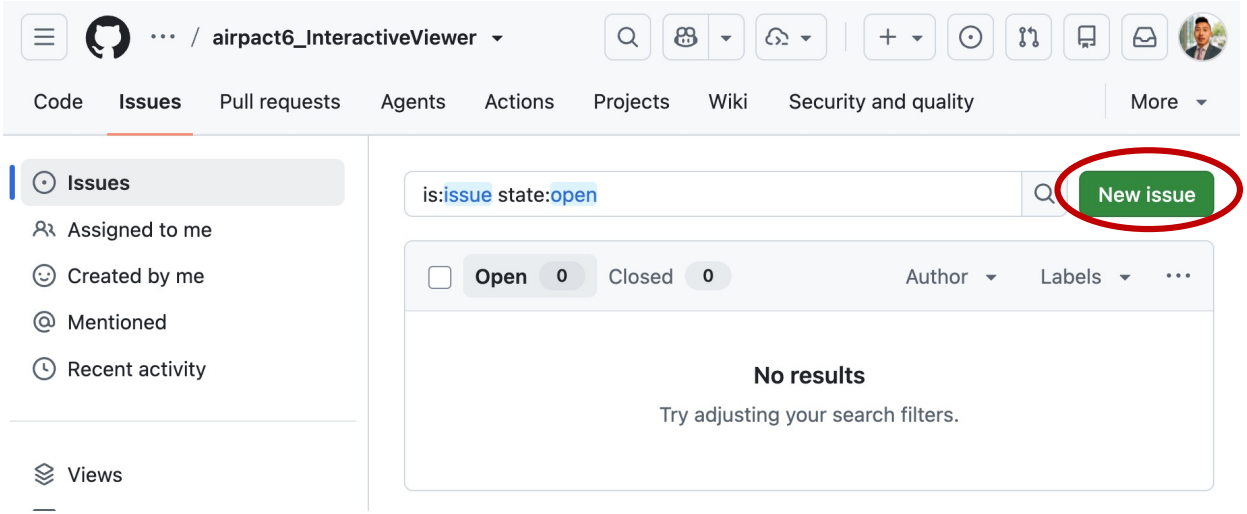
- Currently hosted on Cloudflare – a free cloud web server
- Plan to incorporate it into WSU campus web server – airpact.wsu.edu
- Current features: NRT AirNow overlay, click and query concentration, user-friendly time bar control
- Let us know what features should be changed and what new features should be added

2. AIRPACT6 visualization Prototype is now live

Interactive Map Prototype - https://nw-air-forecast.pages.dev/

Github repo - https://github.com/Jun-Meng/airpact6_InteractiveViewer

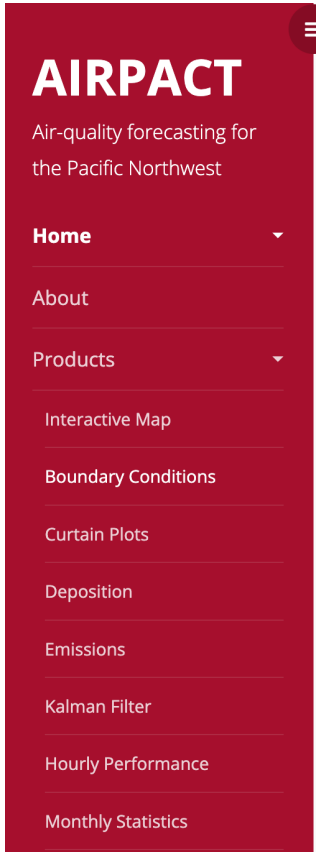
How you can help:

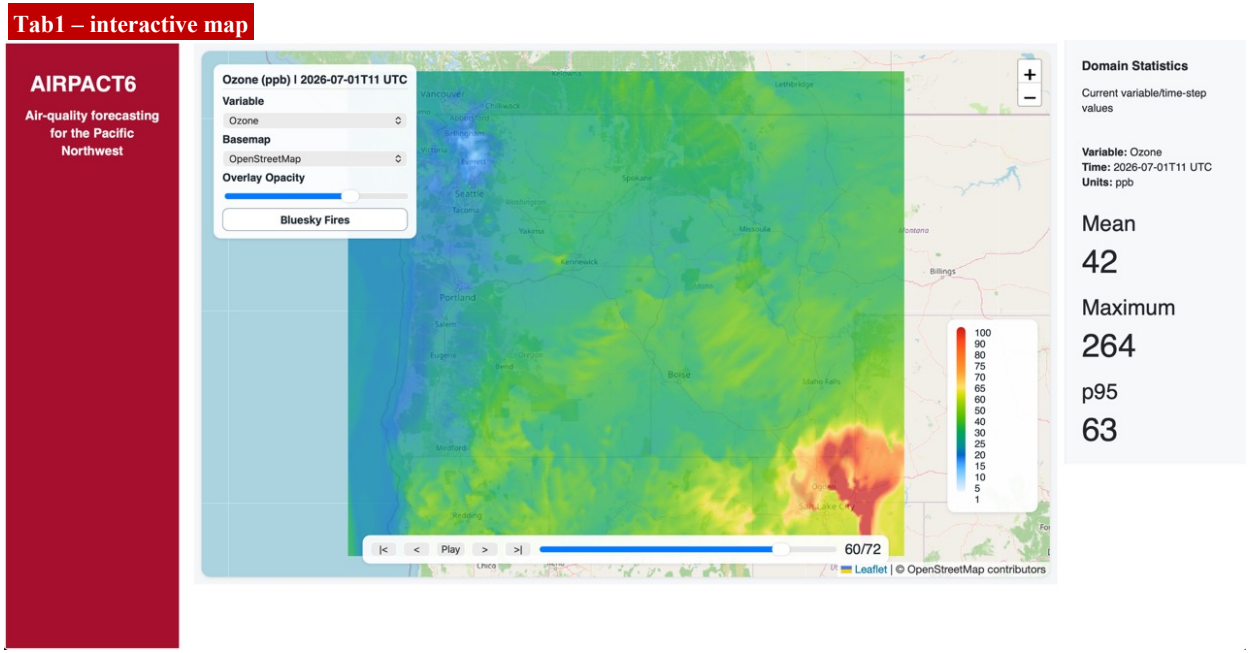


- Bugs
- New feature suggestions
- Comments

3. AIRPACT Website Improvements (not fully delivered)

Current Design → Proposed Design



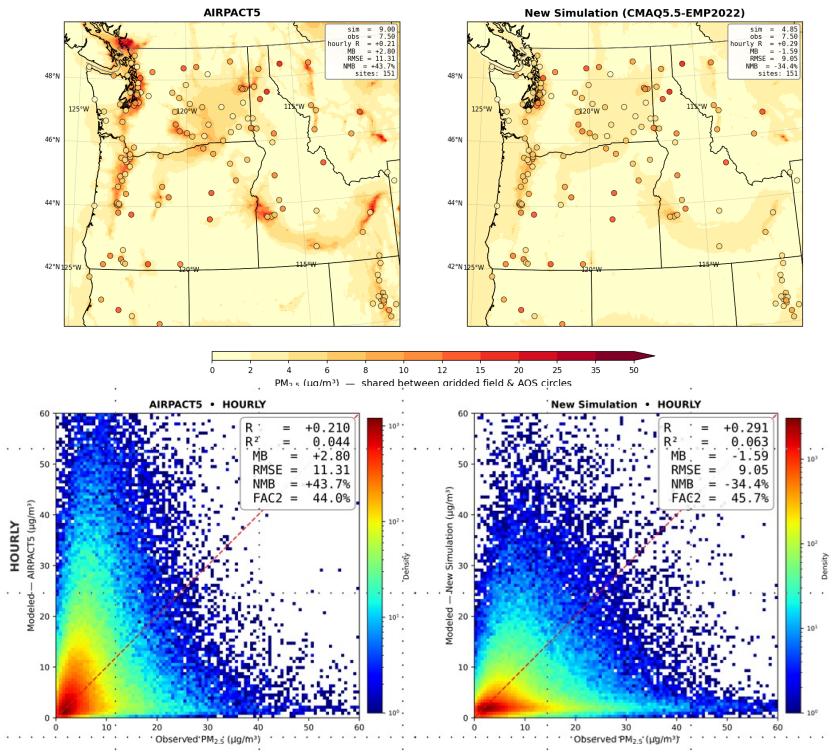


Credits to Von Walden

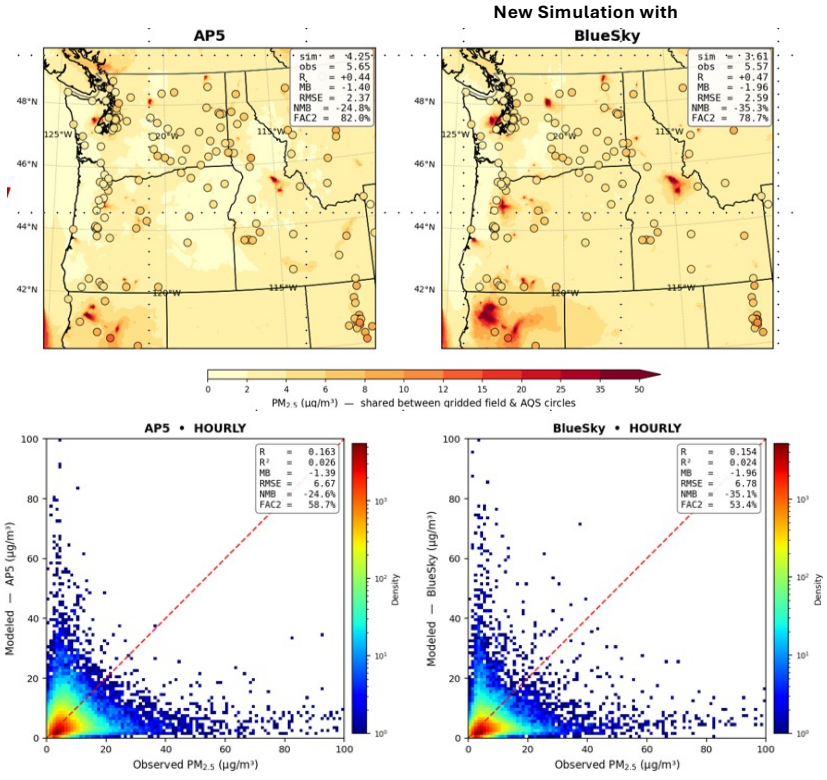
4. AIRPACT6 Summer – Winter Monthly evaluation – preliminary

In winter, AIRPACT6 performs better on a monthly scale, with higher correlation and lower bias.

Winter (January-February 2025)



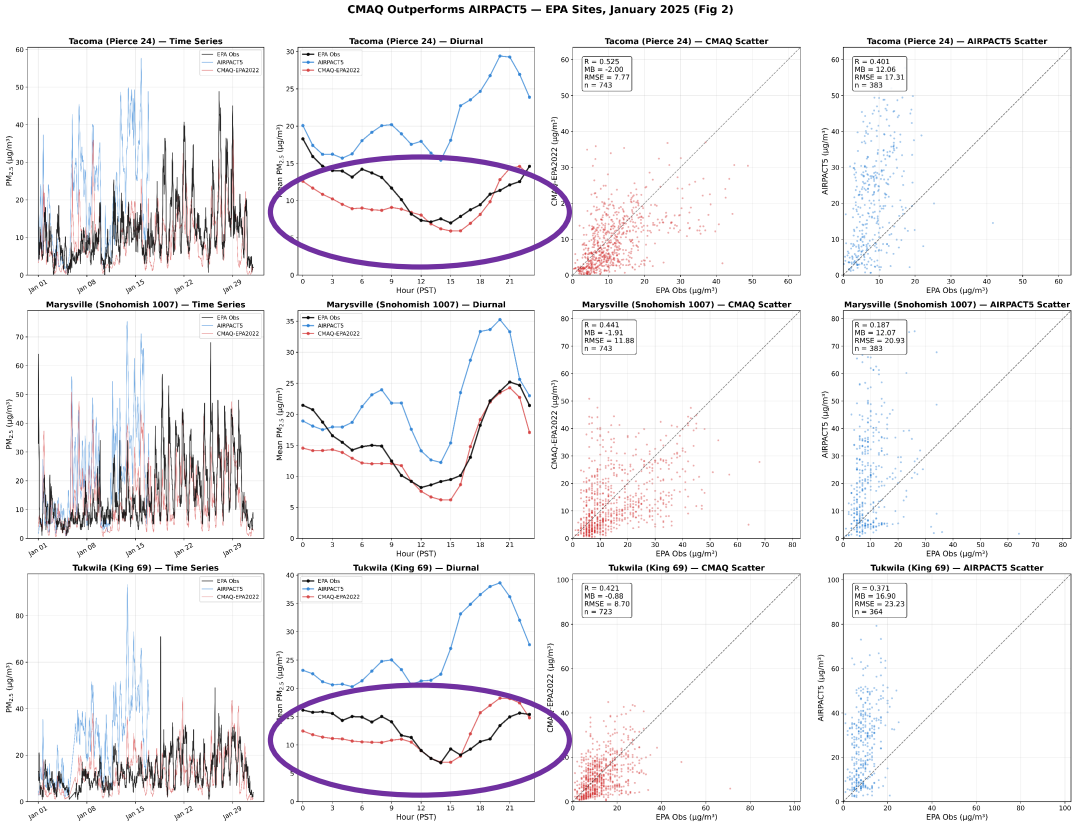
Summer (July 2025)

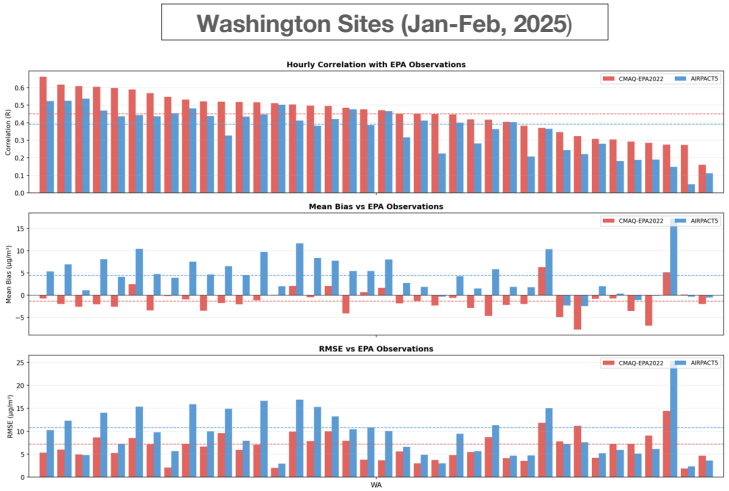


In summer, Bluesky based fire emission did not improve model performance; remains same in hourly resolution.

4. AIRPACT6 evaluation – Winter Month – Jan & Feb 2025

AIRPACT6 substantially improves wintertime diurnal AQ forecast

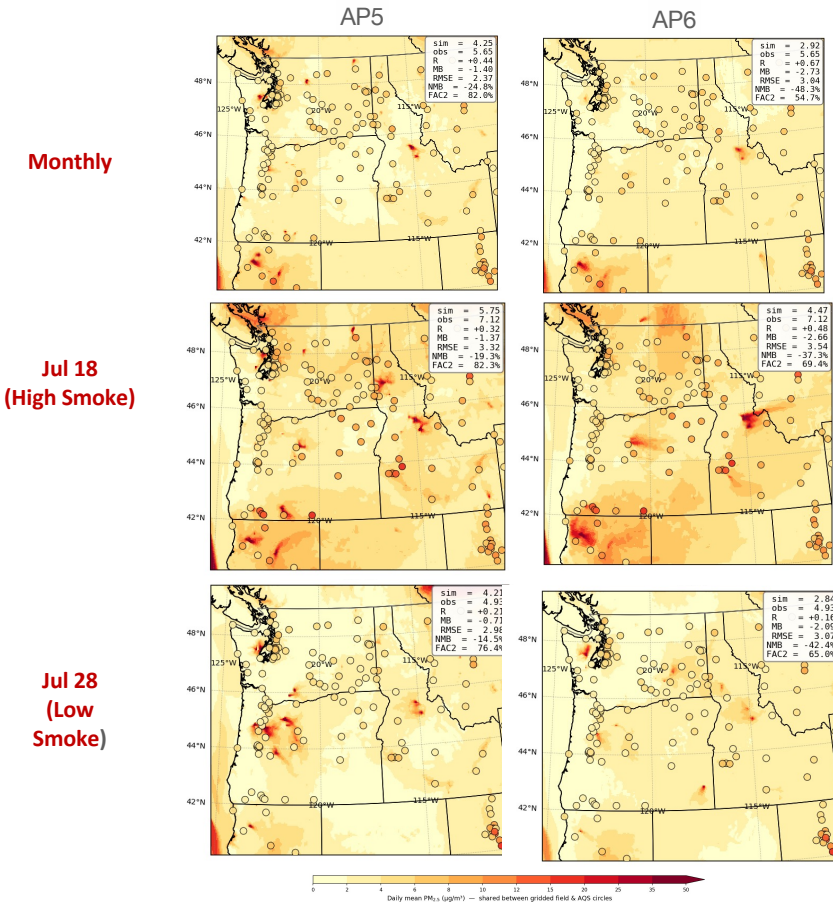


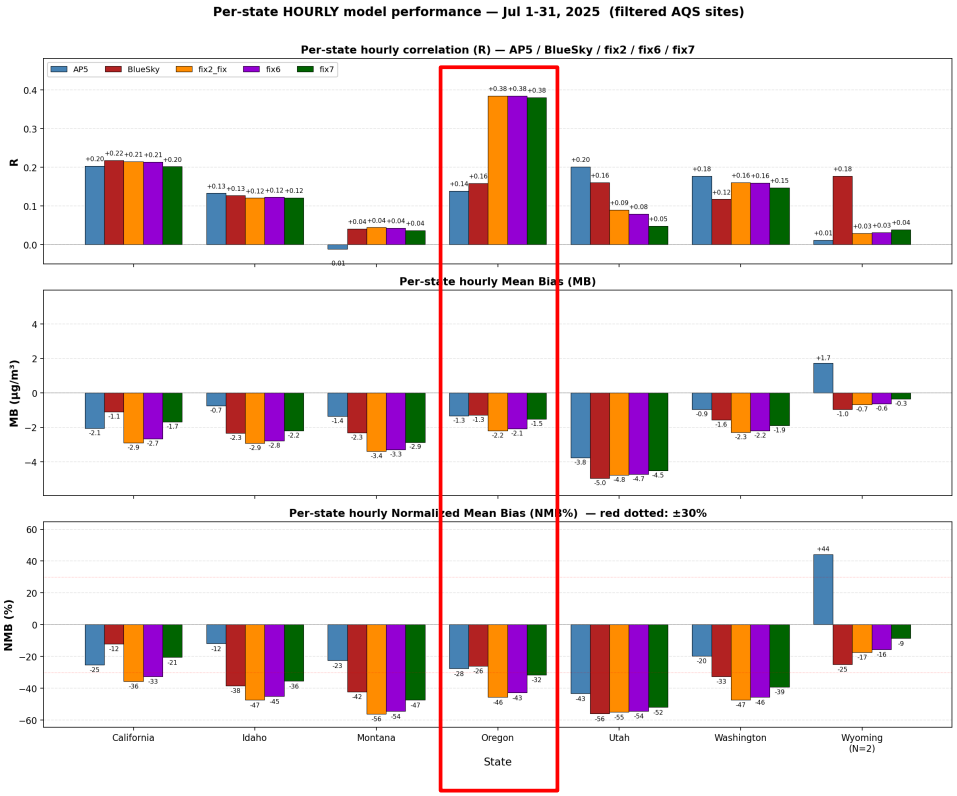


4. AIRPACT6 evaluation - Summer Month – July 2025

Priom will present the details

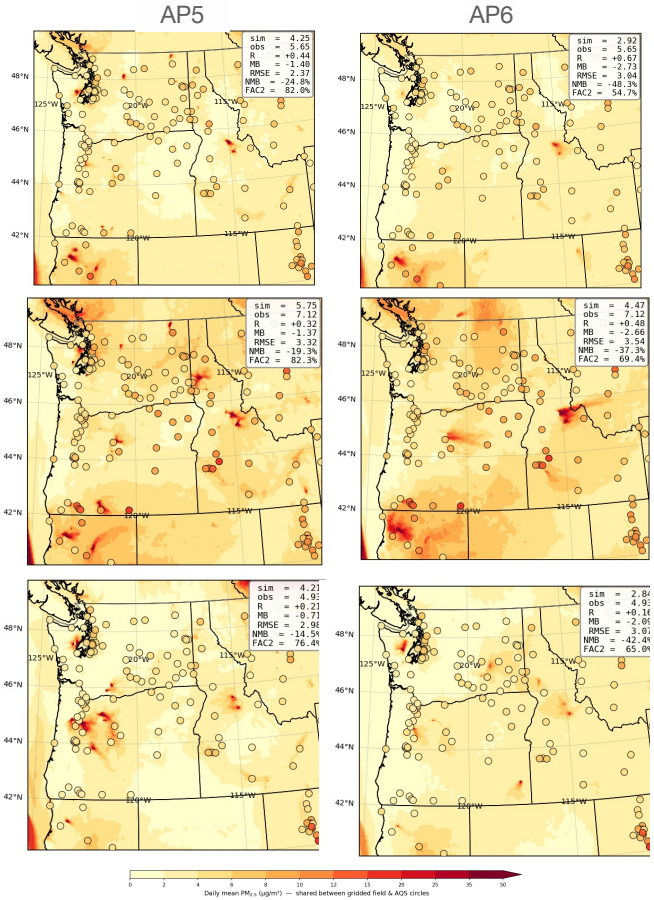
AIRPACT6 with HMS fire emission improves spatial correlation for summertime AQ forecast





4. AIRPACT6 evaluation –

ISSUE: AIRPACT6 underestimates PM2.5 for summertime AQ forecast; most likely due to surrogate issues



Example:

ONROAD

SRGID	Description	Surrogate	Type	Remarks
205	Extended Idle Locations	USA_205_4km.txt	Real	-
239	Total Road AADT	USA_239_4km_proxy.txt	Proxy from USA_308	Proxy from NLCD Low + Med + High
240	Total Road Miles	USA_240_4km.txt	Real	-
242	All Restricted AADT	USA_242_4km.txt	Proxy from USA_240	Proxy from Total Road Miles
244	All Unrestricted AADT	USA_244_4km.txt	Proxy from USA_240	Proxy from Total Road Miles
259	Transit Bus Terminals	USA_259_4km.txt	Real	-
304	NLCD Open + Low Intensity	USA_304_4km.txt	Real	-
306	NLCD Med+High	USA_306_4km.txt	Real	-
307	NLCD Developed Open Space	USA_307_4km.txt	Real	-
308	NLCD Low+Med+High	USA_308_4km.txt	Real	-
508	Public Schools	USA_100_4km.txt (fallback or proxy)	Proxy DROPPED	-

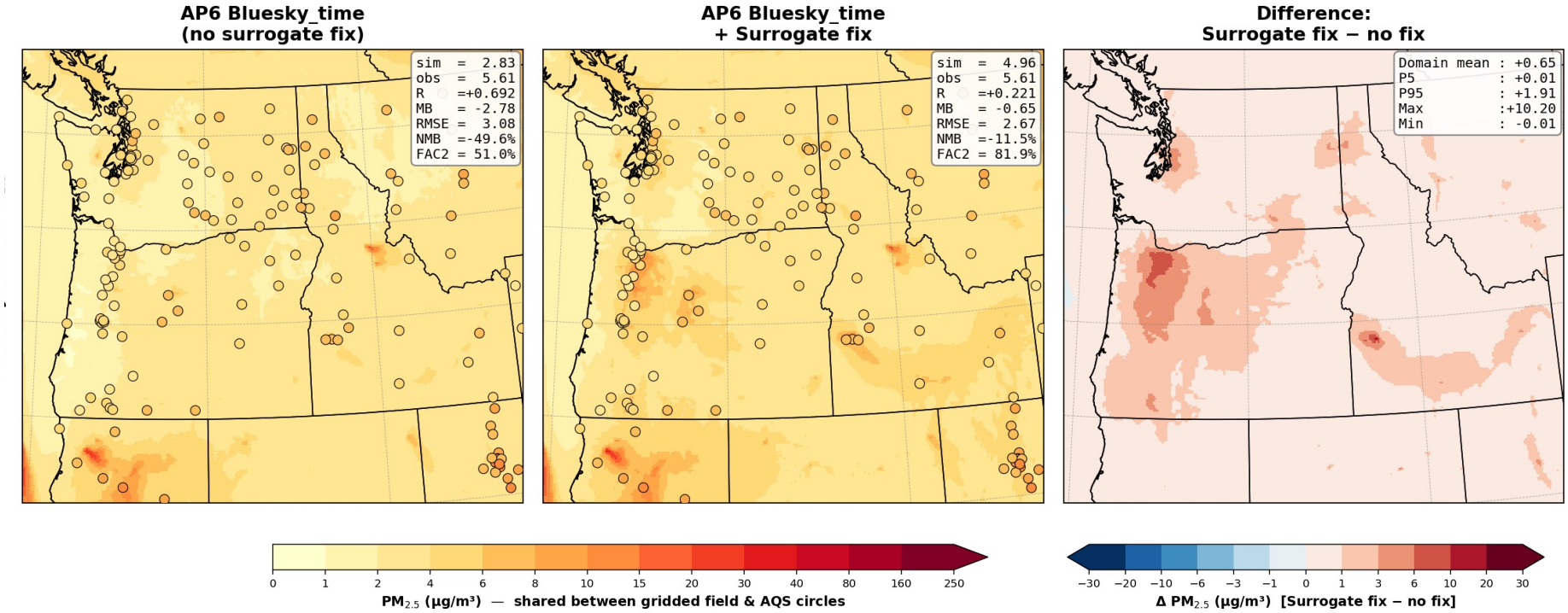
Green rows: EMP2022 required surrogates but not available in AIRPACT 4km grid;

FY27 Proposed Tasks

FY27 PROPOSED TASKS

- **Adjust Anthropogenic Emissions – customized 4km surrogates**
- **Implement the new chemical mechanism (CRACCM) into AIRPACT6 to potentially improve O3 forecast performance**
- **Develop a path forward for a hybrid PM2.5 and O3 bias-corrected forecast product, potentially explore NASA TEMPO products**
- **Display Idaho DEQ's ML forecast on AIRPACT website**

1. AIRPACT6 surrogate issue fix

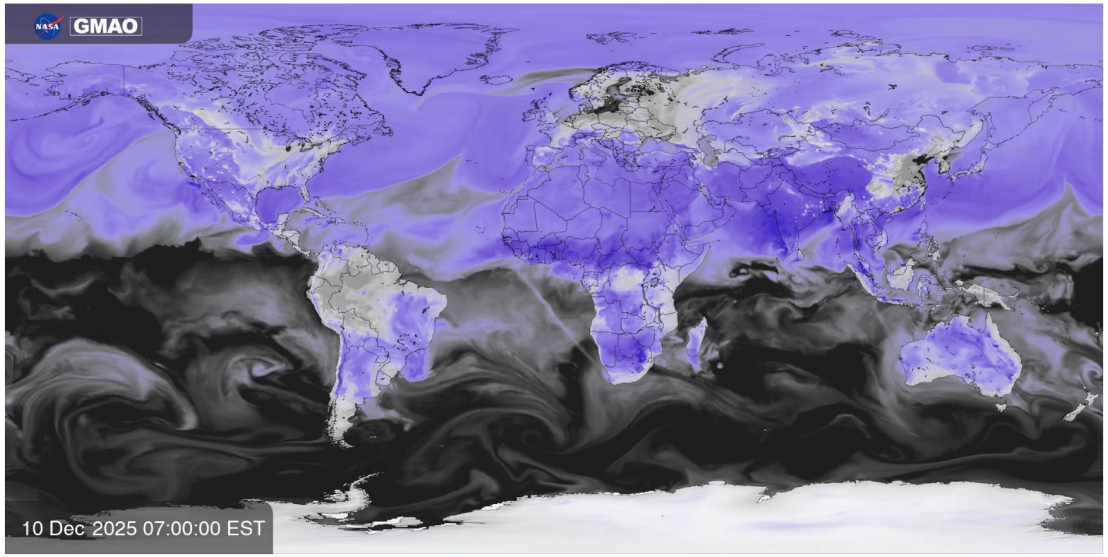


Current testing: New WA Ecology surrogates reduce model bias but degrade spatial correlation.

2. Implement GEOS-CF as Boundary Condition

GEOS Composition Forecasts

The GEOS Composition Forecasting (GEOS-CF) system produces global, three-dimensional distributions of atmospheric composition, including surface pollutants and tropospheric and stratospheric composition. The GEOS-CF products include a near-real time "analysis" segment, where meteorological fields are informed by other GEOS data assimilation systems, and a 5-day "forecast" segment that projects atmospheric composition using internal model physics and GEOS-Chem chemistry.



NASA GMAO

Advantages

- **Higher spatial resolution (~25 km)**
- **More frequent timesteps (every hour)**
- **Potential integration of NASA satellite data assimilation in future releases**

3. Other plans

- Sunset AIRPACT5 operation once we are comfortable with AIRPACT6 anthropogenic emissions

- Monthly emission maps and monthly statistics will be available on the new AIRPACT website

Summary

- AIRPACT6 shows better model performance based on Jan and July 2025 simulations
- HMS FRP based wildfire emission inventory is implemented into AIRPACT6
- AIRPACT6 in forecast mode is running on WSU HPC Kamiak since June 7 2026
- New interactive map is developed; airpact website improvement is ongoing work
- FY27 proposed tasks are underway
- Monthly AIRPACT modeling meeting is ongoing; next meeting will be in September

Thank You!
Any questions?